AQA

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

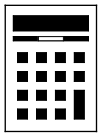
Functional Skills Certificate
FUNCTIONAL MATHEMATICS

Level 2

Tuesday 28 February 2017 Morning Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments
- a copy of the data book (examination) (enclosed).



For Examiner's Use	
Question	Mark
1	
2	
3	
4	
TOTAL	

Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- Evidence of checking is specifically assessed in Questions 1(a) and 4(b). These questions are indicated with a **†**.

Advice
- In all calculations, show clearly how you work out your answer.

Do not write outside the box

Answer **all** questions in the spaces provided.

1 **Tenpin bowling**

There is a **data sheet** for Tenpin bowling.



I am captain of a tenpin bowling team.

Anna

Anna is playing a game against Ben.

†1 (a) Anna has completed two frames of the game.

1		2	
7	/	8	1

Work out her total score after Frame 2

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

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1 (b) Later, Ben has one more frame to play.

1		2		3		4		5		6		7		8		9		10	
4	3	8	/	4	5	**X**		7	2	1	7	5	4	**X**		6	/		
7		21		30		49		58		66		75							

In Frame 10, Ben knocks over

5 pins with his first ball

3 pins with his second ball.

He says,

"My final score is **more than** 115"

Is he correct?
You **must** show your working.

[4 marks]

Question 1 continues on the next page

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Anna has to choose a new player for her bowling team.

She can choose either Jamil or Tom.

Jamil and Tom have each played some games against Anna.

Here are their final scores and results.

Jamil	
Final score	**Result**
155	Lost
147	Lost
216	Won
182	Lost
179	Won
177	Lost

Tom	
Final score	**Result**
191	Won
160	Lost
134	Lost
210	Won
182	Lost
202	Won
159	Lost
146	Lost

Do not write outside the box

1 (c)



I have a greater chance of winning against Anna than Jamil does.

Tom

Is Tom correct?
You **must** show your working.

[3 marks]

1 (d)



On average, my final scores were higher than Tom's.

Jamil

Is Jamil correct?
You **must** show your working.

[4 marks]

14

Do not write outside the box

2 **Sandwiches**



I make and sell sandwiches in my shop.

Amir

2 (a) Amir uses 300 grams of spread to make 25 sandwiches.

He buys the spread in 2.5 kg tubs.

He has no spread left.

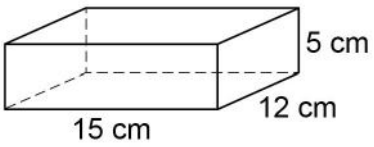
Next week, Amir wants to make 500 sandwiches.

How many tubs does he need to buy?
You **must** show your working.

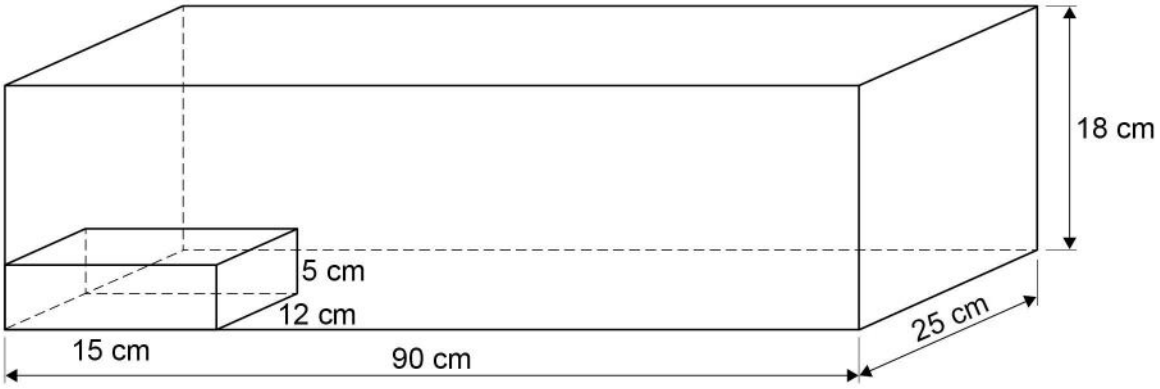
[5 marks]

Do not write outside the box

2 (b) Amir puts the sandwiches in cuboid boxes.



The boxes of sandwiches are packed in cuboid crates.
The boxes are all packed in the same way, as shown.



Work out the **maximum** number of boxes that can be packed in one crate.
You **must** show your working.

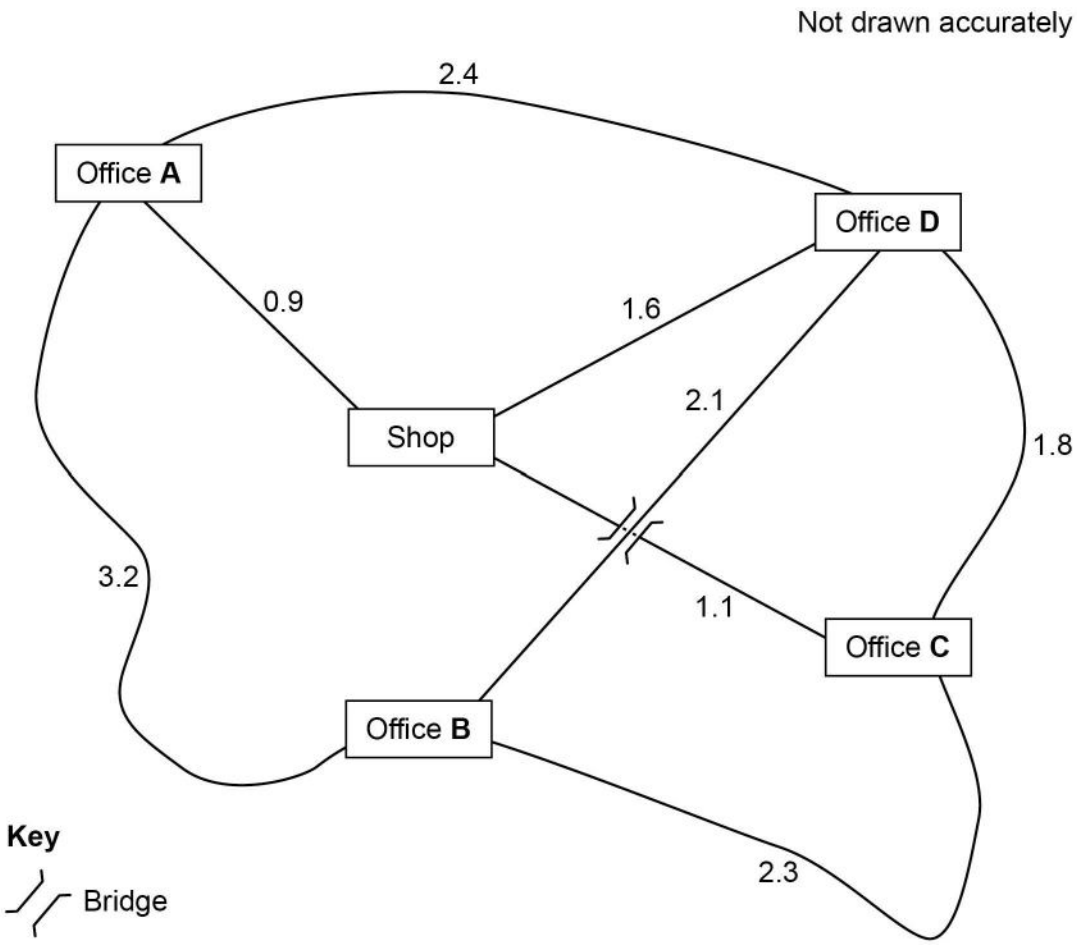
[4 marks]

Do not write outside the box

Amir delivers sandwiches to four offices.

The diagram shows the driving distances between his shop and the offices.

The distances are in miles.



The road between the shop and Office C goes under the bridge.

Do not write outside the box

2 (c) On Monday, Amir drives

Shop ⟶ Office D ⟶ Office C ⟶ Shop

How many miles does he drive?
Circle your answer.

[1 mark]

3.4 4.3 4.5 5.4

2 (d) On Tuesday, Amir drives from his shop.

He visits each of the four offices.

He then drives back to the shop.

Work out the shortest possible route.
Include the route and the total distance he drives.

[3 marks]

13

Do not write outside the box

3 Hairdressing salon



I am opening a hairdressing salon.

Jenny

3 (a) Jenny is designing the salon.

Each sink will need a square space with sides of 50 cm

Each chair will need a circular space with radius 50 cm

She wants

- three sinks against the same wall, at least 1 m apart
- four chairs in a line, each 1 m from one wall
- a rectangular reception desk measuring 1 m by 50 cm
- a rectangular display cabinet measuring 2 m by 1 m
- a rectangular waiting area measuring 2 m by 1.5 m
- the door to be able to open fully.

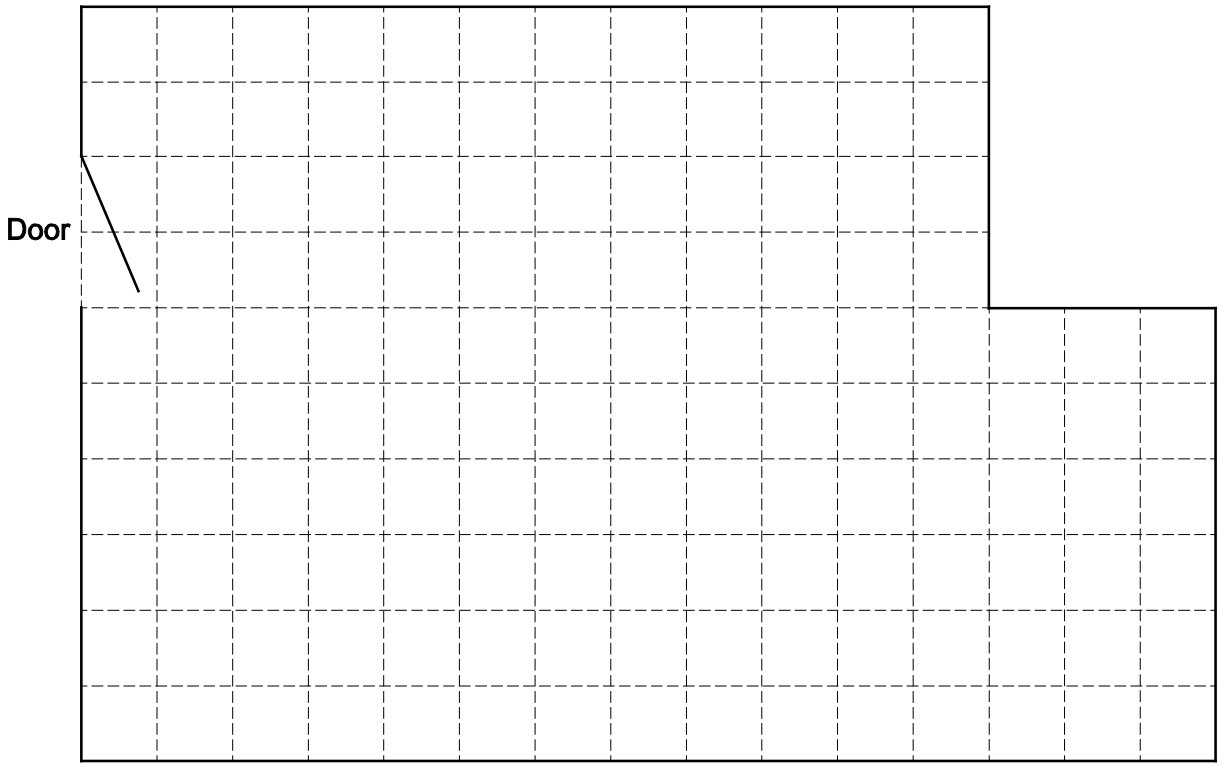
Show a possible design on the scale drawing opposite.

[6 marks]

Do not write outside the box

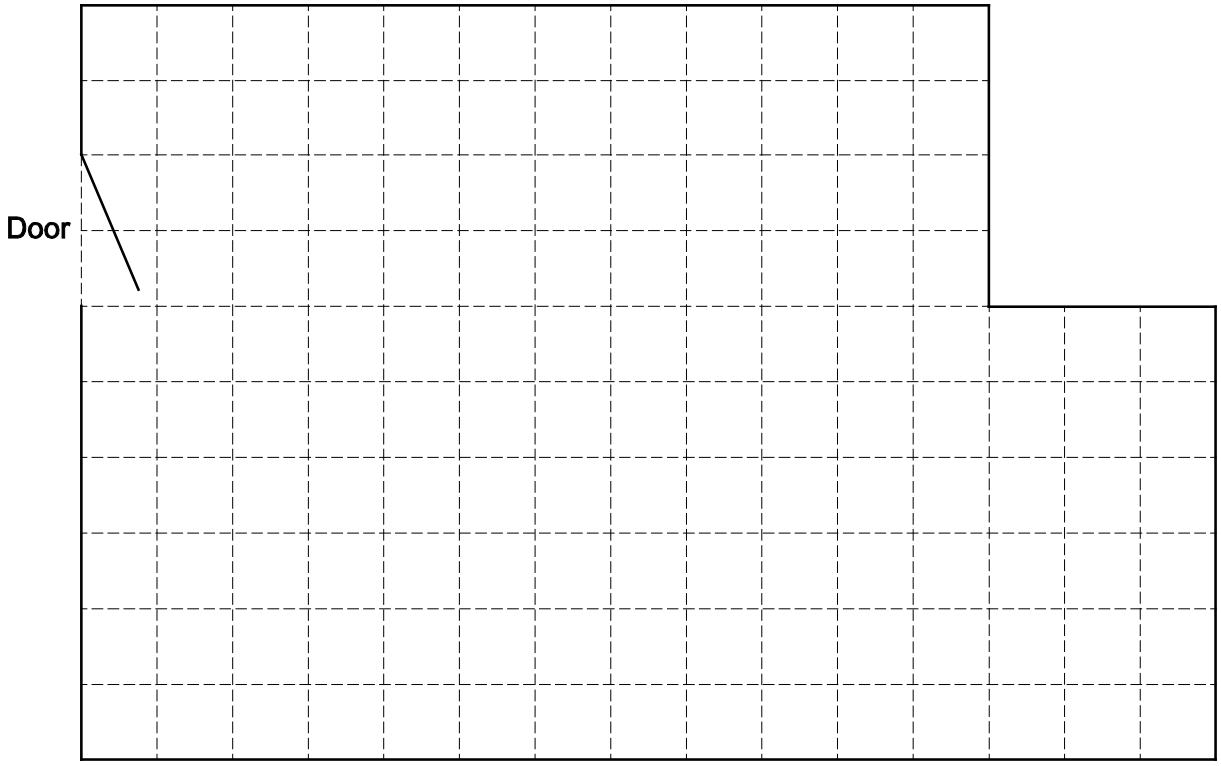
Practise on this diagram.

Scale 2 cm = 1 m



Put your answer on this diagram.

Scale 2 cm = 1 m



Do not write outside the box

Here are the people who work in the salon.

Stylists			
Jenny	Craig	Fay	Mia

Juniors	
Erik	Wendy

3 (b) The salon opens six days each week, from Monday to Saturday.

There are two shifts each day.

From Monday to Friday, two stylists and one junior work each shift.

On Saturday, three stylists and one junior work each shift.

Each week

- Jenny, Mia and Erik each work 8 shifts
- Craig and Fay each work 5 shifts
- Craig does **not** work on Saturday
- Wendy works 4 shifts, all in the afternoon
- each **stylist** has at least 1 full day off.

Make a possible rota for one week.
Use the grids opposite.

[4 marks]

Do not write outside the box

Practise on this grid.

	Morning shift				Afternoon shift			
	Stylist 1	**Stylist 2**	**Stylist 3**	**Junior**	**Stylist 1**	**Stylist 2**	**Stylist 3**	**Junior**
Monday								
Tuesday								
Wednesday								
Thursday								
Friday								
Saturday								

Put your answer on this grid.

	Morning shift				Afternoon shift			
	Stylist 1	**Stylist 2**	**Stylist 3**	**Junior**	**Stylist 1**	**Stylist 2**	**Stylist 3**	**Junior**
Monday								
Tuesday								
Wednesday								
Thursday								
Friday								
Saturday								

Do not write outside the box

3 (c) Here are the prices at the salon.

Cut and blow dry	£33
Cut and colour	£60

In the first week the salon has these appointments

35 for a cut and blow dry
15 for a cut and colour.

The table shows the number of appointments for Craig, Fay and Mia.

	Cut and blow dry	**Cut and colour**
Craig	5	2
Fay	4	3
Mia	16	4

The rest of the appointments are for Jenny.

In the first week Jenny will pay

- 10% of the price of their own appointments to Craig, Fay and Mia
- £980 in other wages
- £325 in other costs.

Work out Jenny's profit in the first week.

[6 marks]

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16

Do not write outside the box

4 **Electricity**

There is a **data sheet** for Electricity.

4 (a) Joe has an electric fire with three settings.

Setting	High	Medium	Low
Power	2 kW	1.5 kW	1 kW

Last week, he used the fire on

High for 6 hours on 1 night
Medium for 5 hours on each of 4 nights
Low for 3 hours on each of 2 nights.

Electricity costs 15.5 pence per unit.

Joe says,

"Last week, the cost of electricity for using the fire was less than £10"

Is he correct?
You **must** show your working.

[6 marks]

Do not write outside the box

Joe has a lamp that uses one light bulb.

†4 (b) An ordinary 100 W bulb

- lasts for 6 months
- costs 89 pence.

How much will Joe pay to buy ordinary 100 W bulbs for his lamp for **4 years**?

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

Question 4 continues on the next page

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4 (c) Joe looks at the cost of buying and using ordinary 100 W bulbs or a low energy bulb for the next **4 years**.

He makes these notes.

The lamp is switched on for 240 hours **each year**.

Electricity costs 15.5 pence per unit.

A low energy bulb

- lasts for 4 years
- costs £13.88
- uses 20% of the number of units of electricity used by an ordinary 100 W bulb.

He says,

"The total cost is less if I use a low energy bulb."

Is he correct?
You **must** show your working.

[8 marks]

END OF QUESTIONS

17

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